

Day 07 - SpaceAgent YouTube 100% Closure

Date: 2026-05-09

Branch: to-knowledge-mc

Lane: SpaceAgent YouTube transcript / metadata research

Status: DEVELOPER-SIDE CLOSED - SERVICE_DOWN

Blocker class: SERVICE_DOWN

Primary blocker: youtube_transcript_connector_not_installed

Closure Decision

Day 07 is closed for developer-side implementation because the YouTube route now reports a canonical proof packet, preserves metadata-only fallback, and refuses to fake transcript-backed claims when the transcript connector is unavailable.

This lane is not GO. The runtime Python context does not have 'youtube_transcript_api' installed, so live transcript extraction is not proven.

What Was Implemented

- Added a canonical SpaceAgent YouTube transcript connector status model.
- Added a proof packet for the YouTube lane with:
 - runtime result,
 - blocker,
 - blocker class,
 - connector identity,
 - no full-video download,
 - no login bypass,
 - no secrets,
 - no raw paths.
- Replaced the vague 'youtube_transcript_connector_not_proven' status at the route level with the exact blocker:
 - youtube_transcript_connector_not_installed
- Preserved legacy blocker compatibility for existing UI/registry references.
- Exposed connector status through:
 - GET /api/gateway/space-agent/youtube/status
 - POST /api/gateway/space-agent/youtube/research
- Kept metadata research available without pretending transcript proof exists.

Files Changed

- src/lib/space-agent-youtube-runtime.ts
- src/lib/space-agent-youtube-connector.ts
- src/lib/space-agent-youtube-connector.test.ts
- src/app/api/gateway/space-agent/youtube/status/route.ts
- src/app/api/gateway/space-agent/youtube/research/route.ts

Routes / Endpoints Checked

- GET /api/gateway/space-agent/youtube/status
- POST /api/gateway/space-agent/youtube/research

UI Behavior

No layout/UI redesign was performed.

Gateway and SpaceAgent surfaces can now consume a truthful YouTube connector status instead of treating a route as live transcript proof.

Service / Runtime Behavior

Runtime dependency inventory:

- python3: present
- youtube_transcript_api: not installed

Loopback-only standalone Mission Control smoke:

- Host bind: 127.0.0.1:3337
- Runtime PID: 16194
- /login: 200
- Unauthenticated GET /api/gateway/space-agent/youtube/status: 401
- Authenticated GET /api/gateway/space-agent/youtube/status: 200
- Authenticated POST /api/gateway/space-agent/youtube/research: 200

Authenticated status result:

- status: limited
- canonical_status: SERVICE_DOWN
- blocker_class: SERVICE_DOWN
- blocked_reason: youtube_transcript_connector_not_installed
- transcript_connector_proven: false

Authenticated research result using public YouTube URL:

- title: Rick Astley - Never Gonna Give You Up (Official Video) (4K Remaster)
- channel: Rick Astley
- publish date: unavailable from oEmbed
- transcript_status: missing
- transcript_segments: 0
- key_claims: 0
- full_video_download_allowed: false
- blocked_reason: youtube_transcript_connector_not_installed

No full video was downloaded. No login, age, region, copyright, or paywall bypass was attempted.

Tests Run

- pnpm test src/lib/space-agent-youtube-connector.test.ts src/lib/space-agent-youtube-scenarios.test.ts src/lib/space-agent-health.test.ts
- pnpm run typecheck
- git diff --check
- pnpm run build
- pnpm test
- node scripts/check-protected-file-invariants.mjs
- staged secret scan
- .env diff check

Validation result:

- Focused SpaceAgent YouTube tests passed.
- Typecheck passed.
- Build passed.
- Full test suite passed: 139 files / 1262 tests.
- Protected-file invariant scan passed.
- Staged secret scan found no secret-like staged values.
- .env diff was clean.

Proof Artifact

The route smoke proved:

- YouTube metadata can be resolved from a public video through the safe metadata path.
- Transcript-backed claims are not generated when the connector dependency is missing.
- The exact runtime blocker is returned.

Remaining Blocker

youtube_transcript_connector_not_installed

Required owner/admin or runtime action:

1. Install the approved transcript connector into the Mission Control runtime Python context:
 - youtube_transcript_api
2. Restart Mission Control if the runtime service does not see the dependency.
3. Re-run:
 - GET /api/gateway/space-agent/youtube/status
 - POST /api/gateway/space-agent/youtube/research
4. Promote only after transcript-backed segments and claims are returned for a public transcript-enabled video.

Safety Confirmation

- No .env changes.
- No secrets printed.
- No auth weakening.
- No public local exposure.
- No full video download.
- No login/copyright/paywall bypass.
- No fake transcript proof.
- No raw local paths exposed in owner-facing output.

Commit / Push

Code commit:

- 9b6f3218045bfa9eacfdcd9275cfccd6d23cd6f3
- Message: feat(space-agent): add youtube transcript closure proof
- Push result: pushed to origin/to-knowledge-mc

Report commit:

- Pending at report creation time.

Rollback

Rollback command:

```
""bash
git revert 9b6f3218045bfa9eacfdcd9275cfccd6d23cd6f3
""
```

Next Day Started

Day 08 - SpaceAgent Firecrawl 100% Closure is automatically started after this Day 07 closeout.